Antiviral Properties of Natural Extracts Against Feline Calici Virus

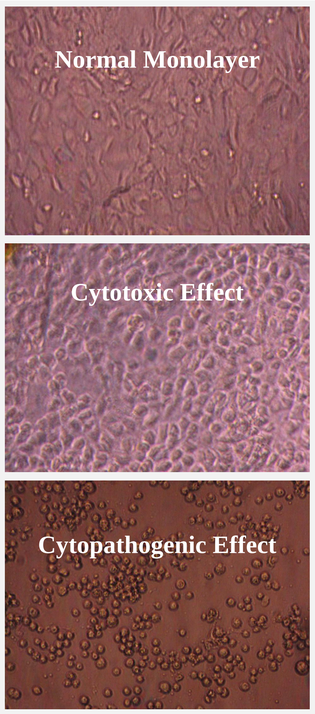
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Introduction

- Due to recent norovirus outbreaks, it has become increasingly important to find effective treatments for this viral infection
- Natural extracts have exhibited antiviral properties in the past(Romeilah 2010; Lee 2012)
- In this study, we explore the antiviral properties of 7 natural extracts using feline calici virus (FCV) as a surrogate to human norovirus

Methods

- **Virus:** FCV-255
- **Tissue culture:** Crandall-Reese feline kidney cells
- **Extracts:** Cloves, Fenugreek, Garlic, Ginger, Habanero pepper, Jalapeno pepper, Onion
- **Cytotoxicity assays:** CRFK inoculated w/ extract only to determine toxicity
- **Cytopathogenicity Assays:** CRFK inoculated w/ virus after treatment w/ extracts
- **Karber's Method:** $TCID_{50}$



Results

Table 1: Anti-FCV activity of natural extracts following pretreatment of CRFK cells with extract

Extract	Dilutions	Log_{10} $TCID_{50}$*		
		30 min	60 min	90 min
Ginger	1/5	6.83	7.50	7.50
	1/10	7.50	7.50	6.50
	1/20	7.50	7.50	6.83
Habanero pepper	1/5	7.17	7.17	7.50
	1/10	7.17	7.50	7.50
	1/20	7.17	7.50	7.50
Jalapeno pepper	1/10	7.17	7.17	7.50
	1/20	7.50	7.17	7.17
	1/40	6.83	7.50	7.50
Onion	1/20	7.17	7.50	7.50
	1/40	7.50	7.50	7.50
	1/60	7.50	7.50	7.50
Cloves	1/30	7.50	7.50	7.50
	1/60	7.50	7.17	7.50
	1/100	7.50	7.50	7.50
Fenugreek	1/50	6.83	7.50	7.50
	1/100	7.17	7.17	7.50
	1/200	7.17	7.50	7.50
Garlic	1/100	6.83	7.17	7.50
	1/150	7.50	7.17	7.50
	1/200	6.83	7.17	7.50

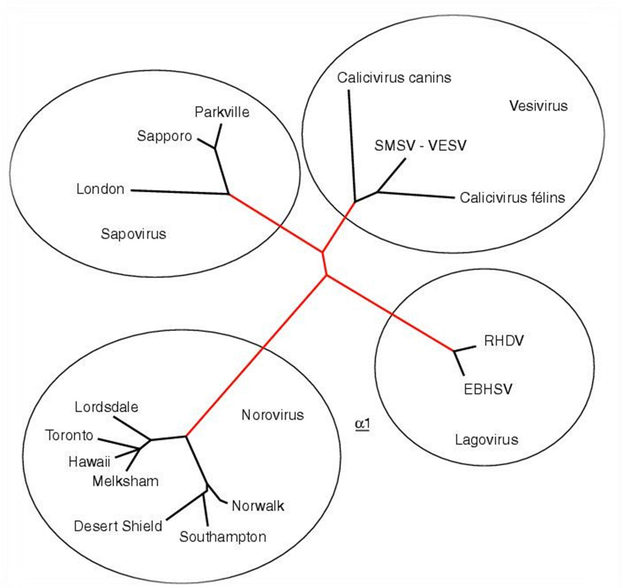
***The intitial titer of FCV was $10^{7.5}$ $TCID_{50}$/mL**

Table 2: Anti-FCV activity of natural extracts following pretreatment of FCV with extracts and co-inoculation

Extract	Dilutions	Log_{10} $TCID_{50}$*		
		30 min	60 min	90 min
Ginger	1/5	7.17	7.50	7.50
	1/10	7.50	7.50	7.50
	1/20	7.50	7.50	6.50
Habanero pepper	1/5	7.50	7.17	6.50
	1/10	7.50	7.17	7.17
	1/20	6.50	6.17	6.50
Jalapeno pepper	1/10	7.50	7.50	7.17
	1/20	7.50	7.17	7.50
	1/40	7.50	7.50	7.50
Onion	1/20	7.17	7.50	6.83
	1/40	7.50	7.50	7.17
	1/60	7.17	7.50	7.50
Cloves	1/30	7.50	7.50	7.50
	1/60	7.50	7.50	7.50
	1/100	7.50	7.50	7.50
Fenugreek	1/50	7.50	7.50	6.83
	1/100	7.50	7.17	7.17
	1/200	7.17	7.17	7.50
Garlic	1/100	7.50	7.50	7.50
	1/150	7.50	7.50	7.50
	1/200	7.50	7.17	7.17

Table 3: Anti-FCV activity of undiluted natural extracts following co-inoculation of extract with virus

Extract	Log_{10} $TCID_{50}$*
Ginger	7.5
Habanero pepper	7.5
Jalapeno pepper	7.17
Onion	7.17
Cloves	6.83
Fenugreek	7.17
Garlic	6.83



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Discussion

- Initial titer of virus = $10^{7.5}$
- Titer of virus after exposure to extract = $10^{6.17\ -7.5}$
- In order to accurately say that a treatment has antiviral properties the titer of virus must be reduced by a minimum of 10^3

Conclusion

- Since only a small amount of virus particles were killed we cannot confidently state that these extracts have antiviral effects against FCV
- Therefore, our initial conclusion is that these extracts have no antiviral effect against FCV, and in turn no antiviral effect against human norovirus

Future Experiments

- Repeat previous experiments entirely
  - re-extract products
  - add + control of known FCV disinfectant
  - statistical analysis

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